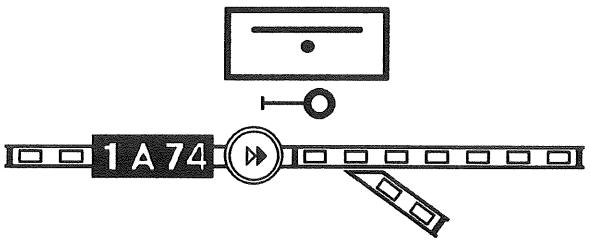
INTRODUCTION TO SIGNALLING

3

THE ELEMENTS OF A SIGNALLING SCHEME



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INTRODUCTION

The signalling system has to perform in a safe manner, but perhaps more importantly, if something goes wrong it also has to fail in a safe manner - for instance if something goes wrong with the controls of a signal, the signal must immediately show a red 'stop' indication to the driver. For a signal to show red when it should be showing green is a nuisance, but not dangerous. This type of failure is known as a *rightside failure*. However for a signal to be showing green when it should be showing red would be very dangerous indeed, and this type of failure is known as a *wrongside failure*. These overall requirements are described as *fail-safe* and are the basis of all signalling systems. Rightside failures are fairly common whilst wrongside failures are extremely rare and each would be the subject of a rigorous enquiry.

THE RAILWAY

The visible parts of the railway that need to be considered consist of the track and the signals.

The track is what the trains run on, and allowance must be made for the lines to diverge and converge. These junctions are known as *points* and a device is needed to move the points from one position to the other to set up the correct path. This is known generically as a *point machine* and is one of the elements that the signalling system has to control.

Signals are placed along the track side and give instructions to the train drivers. Modern signals give their instructions by means of lights, easily visible by day or night. Consequently signals are also one of the elements that the signalling system has to control.

Electrical circuits in the rails, *track circuits*, are used to determine whether a particular section of track is clear of traffic or not.

THE SIGNALLING CENTRE

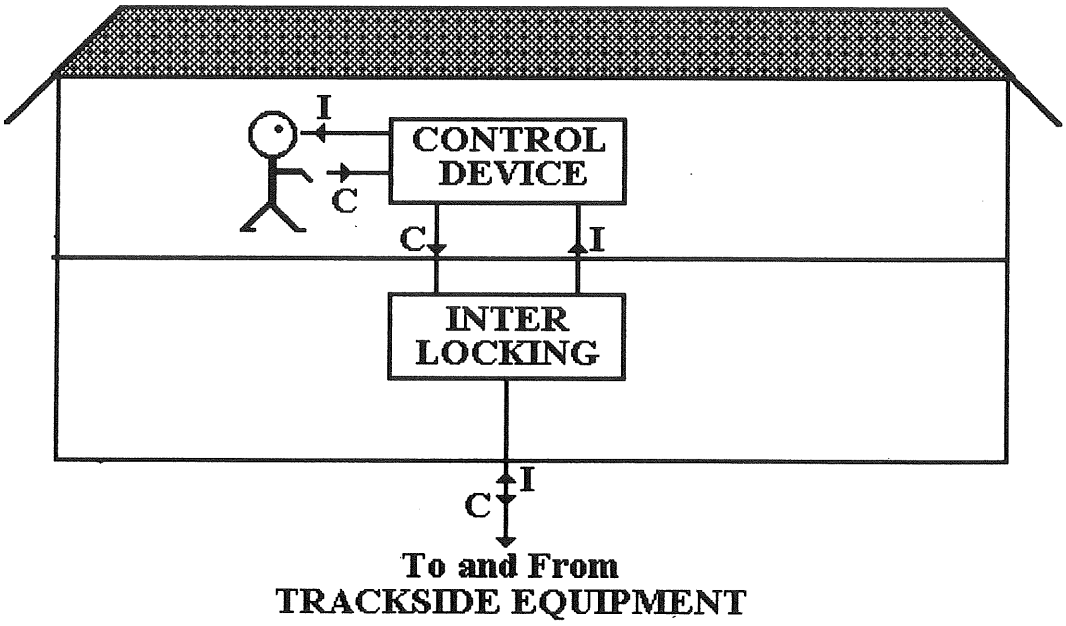
The Signaller

The railway is controlled by one or more *signallers* housed in the *signal box*. Unfortunately, the signaller, being human, cannot be relied on to behave in a fail-safe manner. There is consequently the contradiction of a non-safe signaller trying to control a safe railway. In such a situation the signaller could issue a dangerous set of instructions unless some precautions are taken.

The answer to this problem is to pass the instructions from the signaller through a safety device - the *interlocking* - before they are passed outside to the railway. This interlocking ensures that only safe instructions will be passed on. The interlocking is therefore the most essential element of the signalling system and ensures the safety of the railway, whatever the signaller attempts.

The signaller needs some sort of control device to input his instructions (*controls*) into the interlocking. It is important to have proof from the trackside equipment that any control sent out has been successful, so information is fed back to the interlocking continuously reporting on the condition of the trackside equipment (*indications*). This information - whether each signal is showing a stop aspect or not, which way each set of points are lying, which sections of track are occupied by trains, etc., is indicated for the signaller's benefit on the control device. Both the control device and the interlocking exist in different forms, depending on the age of the equipment.

One possibility, from the early days of signalling, is the control device and the interlocking being combined into one item - *the lever frame*. Individual levers work the points and signals, and the levers are mechanically connected (interlocked) together in such a way that a lever can only be moved by the signaller when it is safe to do so. This apparatus is still in use in places, but will not be considered any further is this document.



The Control Device

Considering the control device, this could be a *control panel* containing all the buttons and switches necessary for the signaller to control his area. A large number of lamps would also be involved indicating back to the signaller the state of the trackside equipment.

Alternatively, in the latest installations a VDU, keyboard and trackerball could be used. The keyboard and trackerball (a mouse lying on its back) would be used to input the controls, whilst maps of the area would indicate information (as described earlier) on the VDU screen.

The Interlocking

For many decades the interlocking was carried out by using *relay logic*. A relay consists of a number of electrical switches that can be opened or closed by passing an electrical current through a coil incorporated in the relay. By suitably interconnecting the relays all the required safety logic can be built up. An interlocking for a medium sized area might well consist of two or three thousand relays, whilst a large signalling scheme could use perhaps 30,000 relays in toto. Relay interlockings require a lot of space and the signaller(s) and control device would normally be on the first floor, above the *relay room*.

If relay technology is used, each control leaving the interlocking, and each indication returning to the interlocking, would involve two wires. So what is shown on the diagram above as one line would in fact consist of many hundreds of wires, grouped together in *multicore cables*. A medium size interlocking might well require 200 controls and 500 indications, resulting in 1400 wires. In fact there would normally be two sets of such multicore cables leaving the signal box, one going along the line in one direction and the other going along the line in the other.

For many years the cost of this cabling limited the size of the area that could be controlled - it was not economical to control junctions at a distance from the signal box, due to the high cost of copper. The development of *Remote Control Systems* overcame this problem. Remote Control Systems enabled a large number of controls and/or indications to be sent over one pair of wires, rather than several hundred. Once this was possible, the area controlled from one place could be expanded considerably, and the *Signalling Centre* (rather than Signal box) was born.

With the ready availability of microprocessors, a solid state interlocking (SSI) system became possible. Before the advent of SSI, electronic devices were not allowed to be used in any safety circuits. An SSI interlocking takes up very little space compared with the relay equivalent and normally a single storey building is quite adequate. All the controls and indications associated with an SSI are passed over a single cable, containing only two wires. However a duplicate cable is installed as a working standby, so what is shown on the diagram as one line would in fact consist of two 2 core cables. Again there would normally be one pair of cables going in each direction.

Power

Electrical power is required, not only to work the equipment in the signalling centre, but also to work the trackside equipment. A *power cubicle* will normally be installed in the signalling centre, and this must provide sufficient power at the various required voltages for all the

signalling equipment in the building. It must also distribute power to the trackside at 650 V, usually in a heavy power cable, with one for each direction. (These power cables are shown as dashed lines in the diagrams below. To keep the diagram simple the internal power distribution is not shown.)

The power cubicle is supplied with power by the local electrical distribution company. The supply needs to be very secure, and if possible should be dual-sourced from two different electrical substations. Alternatively, a diesel standby generator set may be installed, or possibly an emergency supply obtained from the traction supply on an electrified railway.

The Train Describer

On a large scheme a signaller might well have several trains under his control at any one time. The positions of trains are indicated to him from information returned from the track circuits. However it will be difficult to remember which train is which. The *Train Describer* gives the signaller additional information by indicating the number of each train. Each train on a given day has a unique number, and this number is shown in the working timetable (but not in the public timetable). If these numbers are shown to the signaller he will be in no doubt as to which train is which and will be able to route them appropriately.

The Train Describer receives information as to the position of trains from the interlocking (from track circuit information), and then 'associates' a train number with each.

The Train Describer does not count as a fail-safe device. If it goes wrong and shows an incorrect number to the signaller, then all that will happen is that the train might be routed the wrong way by the signaller. Inconvenient, but not dangerous as the interlocking would still block any dangerous moves.

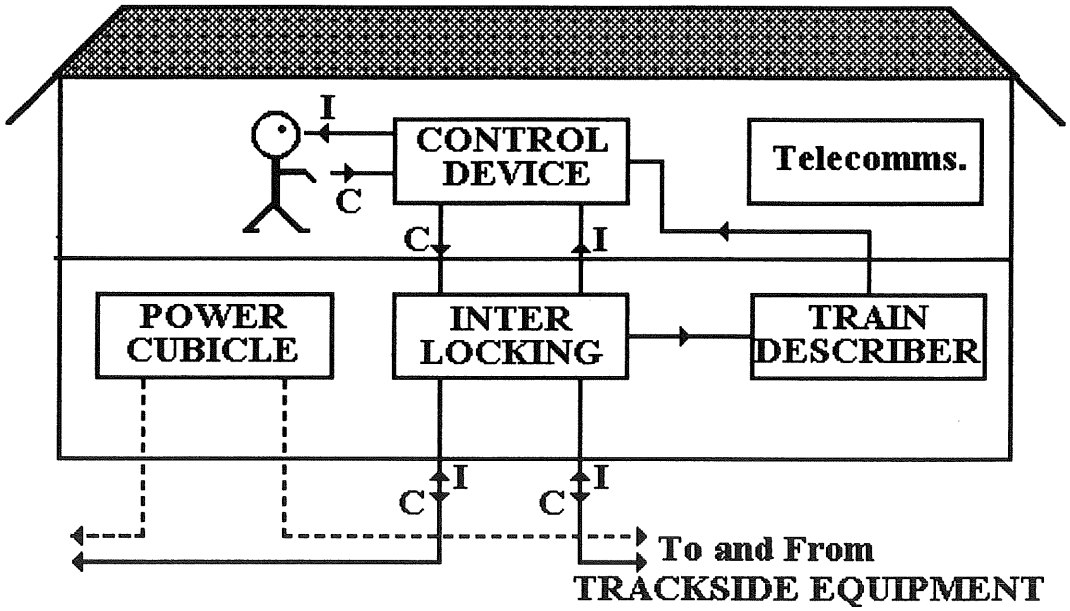
In a modern installation the control device and the Train Describer might well be combined together in an *Integrated Electronic Control Centre*. This incorporates all the non-vital parts of the control centre equipment in one package. Other advanced features such as automatic route setting direct from the time table might also be incorporated.

Telecommunications

The railway needs a comprehensive telecommunications system linking signalling centres, railway offices, workshops etc., for both voice and data. In addition, there are extra facilities required by the signalling system. On modern schemes, every signal has a *signal post telephone*, so that a driver stopped at a signal can talk to the signaller to ascertain the situation, and be given instructions as appropriate. In a lot of areas direct radio communication between the signaller and the drivers is also provided. In addition, phones are provided in the areas of points, to allow easy communication for testers and maintainers.

Summary

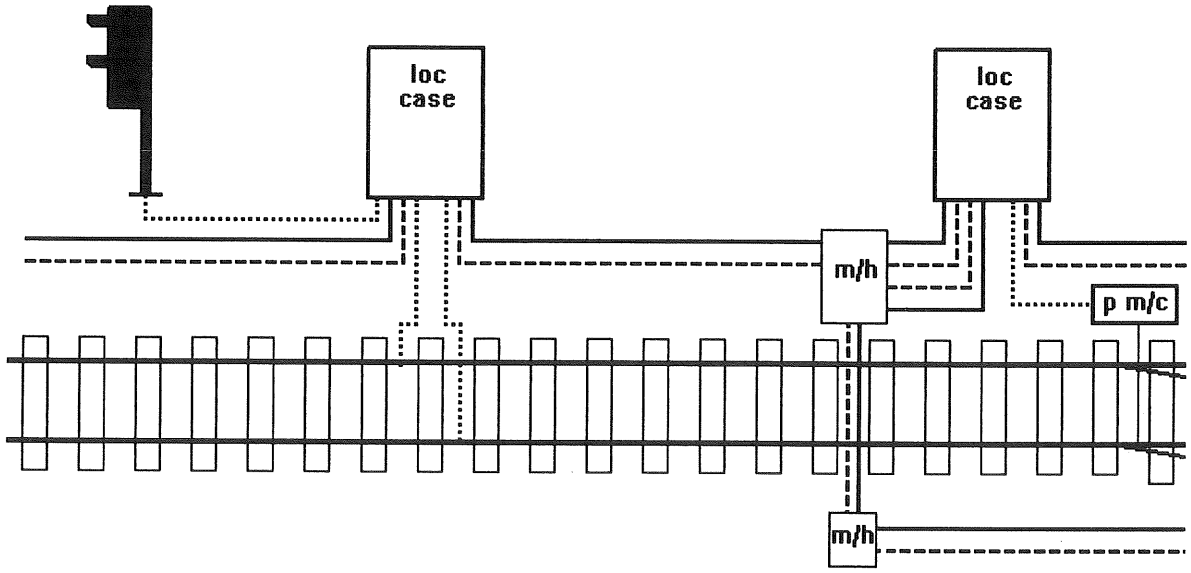
The diagram below summarises in general terms the items described above relating to the signalling centre.



THE TRACKSIDE EQUIPMENT

The cable route

The cables to and from the signalling centre are housed in a *cable route* installed alongside the railway. This route usually consists of concrete troughing laid in the ground on one side of the track or the other. The troughing is supplied with loose lids that are placed on top of the troughing once the cables are in place. If the cable route has to actually cross the railway somewhere a formal *undertrack crossing* has to be constructed. This consists of a manhole (m/h in the diagram) each side of the line, connected together by a small tunnel. These crossings are expensive to install so their number should be kept to an absolute minimum. The cable troughing would contain both the signalling cables (drawn as a solid line) and the power cables (drawn as a dashed line).



Apparatus Cases

The final interfaces to the trackside equipment are housed in *Apparatus Cases*, also known as *Location Cases* or frequently just *Locs*. These are cupboards, about 2 metres high, mounted adjacent to the cable route close to the signals and point machines. The signalling and power cables are terminated in the loc, which also houses power equipment and the final control equipment for the trackside items. Controls are received in the signalling cable from the interlocking and indications are sent in the signalling cable back to the interlocking. Finally there will be short cables from the loc to the signals, point machines and to the rails. These cables are known as *tail cables* and are shown dotted in the diagram above. It is not necessary to construct an undertrack crossing if a tail cable needs to cross the track, it may be laid across

the ballast protected by a plastic pipe. If a loc area is in a complex position more than one cupboard may be required to hold all the necessary equipment. However, if more than three cupboards are required a cheap building of the 'portacabin' variety will be installed instead if space allows. These buildings are technically known as *Relocatable Equipment Rooms*.

Level Crossings

If a railway and a road cross each other on the level, then a *level crossing* is involved. There are a large number of varieties, some with full barriers, some with half-barriers and some with no barriers all. Depending on the levels of both road and rail traffic, some barriers may be allowed to be worked automatically by the approach of trains, whilst others are manually controlled at the crossing, or from a distance using Closed Circuit Television systems.

Automatic Train Control

To improve safety it is possible to pass information from the track to the train and give the drivers additional information to assist their task, particularly important as train speeds increase. One system, the *Automatic Warning System*, gives the driver audible warnings on the approach to each signal, indicating whether the signal is showing a full proceed indication or not.

With a more advanced system, *Automatic Train Protection*, an on-board computer calculates continuously the maximum speed at which `1 the train may proceed, from information picked up as the train moves along the line. If the driver ignores this maximum limit, then firstly an audible warning is given, and if this too is ignored, the brakes are applied automatically, whether the driver likes it or not. With this type of system, trackside signals could be dispensed with.

Hot Axle Box Detectors

It is necessary to monitor the bearings of axles of vehicles at strategic points. If a bearing was to seize up, then the axle would stop rotating, resulting in a 'flat' being worn on the wheel, and possible damage to points and crossings. *Hot axle box detectors* are mounted on the track on the sleeper ends, and measure the temperature of every passing axle end. The number of passing axles are counted, so if a 'hot box' is detected, a message is passed to the controlling signal box indicating which axle of the train is affected. Hot box detectors would normally be positioned on the approach to suitable facilities for taking the train out of service and isolating the faulty vehicle.